

Q1. Entry speed

The descent is frictionless and the total drop is h , so by conservation of energy

$$mgh = \frac{1}{2}mv_0^2 \implies \boxed{v_0 = \sqrt{2gh}}.$$

Q2. Speed before the wall

On the rough horizontal track the kinetic friction is $\mu_k mg$, giving a deceleration $\mu_k g$ over distance L_1 :

$$v_1^2 = v_0^2 - 2\mu_k g L_1 = 2g(h - \mu_k L_1) \implies \boxed{v_1 = \sqrt{2g(h - \mu_k L_1)}}.$$

Q3. After the collision

The incoming velocity is $(v_1, 0)$ along $+x$. For a smooth wall whose surface makes angle θ with $+x$, the tangential component $v_1 \cos \theta$ is preserved and the normal component $-v_1 \sin \theta$ reverses. Recombining,

$$\mathbf{v}' = v_1 \cos \theta \hat{\mathbf{t}} + v_1 \sin \theta \hat{\mathbf{n}} = v_1 (\cos 2\theta, \sin 2\theta),$$

so

$$\boxed{v_x = v_1 \cos 2\theta, \quad v_y = v_1 \sin 2\theta},$$

with v_1 from Q2. (For $\theta = 90^\circ$ the cart bounces straight back; for $\theta = 45^\circ$ it leaves at 90° to the track.)

Q4. Sliding to rest

The collision is elastic, so the speed is still v_1 ; the cart now slides on the plane with friction $\mu_k g$. The stopping distance is

$$d = \frac{v_1^2}{2\mu_k g} = \frac{2g(h - \mu_k L_1)}{2\mu_k g} = \boxed{\frac{h}{\mu_k} - L_1}.$$

Q5. Maximum acceleration

On the frictionless quarter-circle the acceleration has magnitude

$$a = \sqrt{a_t^2 + a_c^2}, \quad a_c = \frac{v^2}{h}, \quad a_t = g \cos \varphi,$$

where φ is measured from the top of the circle. Using $v^2 = 2gh(1 - \cos \varphi)$,

$$\frac{a^2}{g^2} = \cos^2 \varphi + 4(1 - \cos \varphi)^2,$$

which is maximised at $\varphi = \pi/2$ (the bottom), where $a_c = v_0^2/h = 2g$ and $a_t = 0$. The vertical track gives only g and the horizontal track only $\mu_k g < g$, so

$$\boxed{a_{\max} = 2g}.$$

Q6. Total path length

The path consists of the quarter-circle arc, the straight run to the wall, and the post-collision slide:

$$L_{\text{tot}} = \frac{\pi}{2}h + L_1 + d = \boxed{\frac{\pi}{2}h + \frac{h}{\mu_k}},$$

using $d = h/\mu_k - L_1$ (the L_1 terms cancel).

Q7. Total displacement

Take $+x$ along the track toward the wall, $+y$ horizontal perpendicular to the track, and $+z$ vertically upward. The release point is at $(-h, 0, h)$ (the top of the quarter-circle), and the final stopping point is at

$$(L_1 + d \cos 2\theta, d \sin 2\theta, 0).$$

Therefore the displacement vector is

$$\boxed{\Delta \mathbf{r} = (L_1 + d \cos 2\theta + h, d \sin 2\theta, -h)},$$

with $d = h/\mu_k - L_1$, and its magnitude is

$$\boxed{|\Delta \mathbf{r}| = \sqrt{(L_1 + d \cos 2\theta + h)^2 + (d \sin 2\theta)^2 + h^2}}.$$